Question 5: Neural Network Pruning (10 points)

Assume we want to prune a network with 1 million parameters to 80% sparsity (i.e. we want *at most* 200K parameters in the pruned model), using iterative magnitude pruning with rewinding to epoch 5. The original training pipeline has 200 epochs and we prune 20% of the remaining weights at every pruning iteration.

Note: after we get the desired sparsity, we rewind to epoch 5 and train the model to convergence.

D

1. **(3 points)** How many times do we need to rewind to get the final pruned model?
   - A. 1
   - B. 2
   - C. 4
   - D. 8
   - E. 10

need $0.8^n \leq 0.2$

$0.8^8 \approx 0.168$

C

2. **(4 points)** How many epochs of training do we need in total (including the original training pipeline) to get the final pruned (and trained) model?
   - A. 980
   - B. 1600
   - C. 1760
   - D. 1800

$200 + 8 \times 195$

BCD

3. **(3 points)** How much theoretical speedup (measured by FLOPs before pruning divided by FLOPs after pruning) is possible? Select all that applies.

   Hint: Consider the VGG example mentioned in class.
   - A. No speed up.
   - B. 2× speed up.
   - C. 5× speed up.
   - D. 8× speed up.

remove FC weights

remove conv weights.